META AI HACKATHON 2026 · MODEL ECOSYSTEM TRACK · CEPHALOMETRIC TELEMETRY · ZERO-IDENTITY-SWAPPING · CONFIDENTIAL ENGINEERING DOSSIER

Biometric Hair AI

Comprehensive Technical Audit, Architecture Breakdown & Unresolved Problem Solving Dossier

Prepared for technical advisory consultation with Claude (Anthropic) · Complete forensic analysis of implementation progress, production pipelines, WebGL/2D canvas tracking, and lingering failure modes.

REPOSITORY & BUILD TARGET
`c:\biometric_hair` · Expo 58 / React Native

ACTIVE META ECOSYSTEM
SAM 3.1 · Muse Wonder · Muse Image · Llama 3.2 Vision · SeamlessM4T · AudioCraft

DATE & STATE
November 4, 2026 · Pre-Submission Forensic Audit

1. Project Vision & The "Zero-Identity-Swapping" Mandate

Biometric Hair AI is an enterprise-grade biometric hairstyling and cephalometric recommendation system built for the *Meta AI Developer Hackathon*. Traditional generative makeover applications (such as FaceApp, Lensa, and generic Stable Diffusion / ControlNet pipelines) suffer from an unacceptable clinical flaw: **"Identity Drift"**. In generating a new hairstyle, existing models alter the user's bone structure, reconstruct their facial thirds, smooth away distinctive features, modify ethnic skin pigmentation, and replace the user's authentic facial identity with a synthetic stranger.

> **The Core Anthropometric Thesis**
> Hairstyles should not be applied as cosmetic filters. A haircut is a **volumetric compensatory architectural counterweight** designed to balance neoclassical cephalometric proportions:
>
> - **Neoclassical Farkas Facial Thirds:** The canonical facial profile divides into three equal 33.3% segments: Upper (Trichion hairline to Glabella brow line), Middle (Glabella to Subnasale nose base), and Lower (Subnasale to Gnathion chin tip).
> - **Facial Index (Morphological Breadth):** Measured as `(Morphological Facial Height / Bizygomatic Breadth) × 100`.
> - **Golden Ratio Harmony (φ = 1.618):** Scoring the deviation between facial width and longitudinal cranial balance.

Our mandate requires **Zero-Identity-Swapping**: The user's face (forehead, brows, eyes, glasses, nose, cheekbones, lips, jaw, and chin) must remain **100% bit-for-bit mathematically immutable**, while the hairstyle is dynamically composited, volumetrically scaled, and shaded to sit seamlessly on the cranial vault.

2. End-to-End Meta AI Ecosystem Integration

To maximize technical score in the *Meta Model Ecosystem* category, the application fuses eight specialized Meta AI models and sensory pipelines into a unified, zero-latency mobile workflow:

META AI MODEL	SERVICE FILE	CORE RESPONSIBILITY & OPERATIONAL MECHANICS
Meta SAM 3.1 *Segment Anything*	`src/services/perceptionService.ts`	**Point-Prompted Scalp Masking:** Injects positive prompt points strictly in the cranial vault above Landmark 10 (Trichion). Injects negative facial protection prompts at Landmarks 9, 168, 234, 454, and 152. Enforces: `Mask = SAM31_Hair_Mask AND NOT(Face_Oval_Polygon)`.
Meta Llama 3.2 Vision *11B Vision Instruct*	`src/services/perceptionService.ts`	**Reference Inspiration Parser:** Accepts an optional user-uploaded photo from gallery. Extracts texture (straight, wavy, curly, textured), volume (low, medium, high), parting style (curtain, side, swept back), and passes parameters to Muse Wonder.
Meta Muse Wonder *Cephalometric Reasoner*	`src/services/synthesisService.ts` `src/services/basetenService.ts`	**Volumetric Architecture Planner:** Fuses MediaPipe Farkas thirds and facial index with Llama inspiration. Computes morphological classification (Oblong, Oval, Round, Square), suitability score, and prescriptions for crown elevation, side width, and fringe drop.
Meta Muse Image *Photorealistic Synthesis*	`src/services/synthesisService.ts` `src/services/photorealisticHairSynthesis.ts`	**Mask-Bounded Inpainting:** Generates photorealistic hair texture within SAM 3.1 boundary using negative prompts ("*no cartoon, no generic AI artifacts, photorealistic, professional salon photography*"). Executes mathematical blit pass to restore raw facial pixels.
Meta Chameleon *Multimodal Lookbook*	`src/services/synthesisService.ts`	**Interleaved Directives:** Generates multimodal style guide cards blending high-fashion editorial imagery with technical directives (`01 · CROWN ELEVATION`, `02 · TEMPORAL CONTOUR`, `03 · FOREHEAD FRAMING`).
Meta Muse Spark / Llama 3.1 *Salon Concierge*	`src/services/interactionService.ts`	**Interactive Salon Assistant:** Chat interface answering user queries regarding product hold, maintenance routines, and styling techniques based on the user's specific facial morphology.
Meta SeamlessM4T v2 *Multilingual Translation*	`src/services/interactionService.ts`	**Real-Time Internationalization:** Wraps all Concierge and Barber Directives in a translation layer supporting English, Spanish (Español), Mandarin (中文), and French (Français).
Meta AudioCraft / AudioGen *Sensory UX Acoustics*	`src/services/interactionService.ts`	**Procedural Audio Synthesis:** Synthesizes audio via Web Audio API: 240Hz-480Hz dual-sine harmonic drone on biometric initialization and mechanical camera shutter on snapshot capture.

3. Visual Feature Showcase & UI Walkthrough

Below is the complete photographic and telemetry audit of every feature implemented across the application lifecycle.

3.1 Before Image Capture: Live Camera View & Biometric HUD

The entry screen is camera-first, operating at a target 60 FPS. MediaPipe tracks 468 landmarks in real time, overlaid with the cephalometric golden ratio reticle and native iOS control bar:



Figure 1A: Live Camera Feed with Textured Crop Lens

Real-time cephalometric HUD tracking trichion, glabella, and temporal crests. Live 2D/3D hybrid hair preview rendered at 60 FPS with Euler pitch/yaw/roll head pose integration.



Figure 1B: Live Tracking with Curtain Fringe Lens

Golden Ratio φ reticle active. Real-time bitemporal span tracking between Landmark 127 and 356, maintaining accurate lateral hair width.



Figure 1C: Refined Pearl White iOS Luxury Theme

Native light theme (#F7F8FA) with delicate grey borders (#DDE1E6). Bottom action panel featuring UPLOAD, primary CAPTURE & ANALYZE shutter, and STYLES drawer.

Native Camera Architecture Highlights

- Hardware MediaStream decoder with dynamic aspect ratio sync.
- Dual-mode biometric pipeline: MediaPipe WASM with zero-latency YCbCr skin chrominance fallback.
- 60 FPS spring physics damper (K=35.0, C=6.5) preventing jitter during rapid head movement.
- Instant language toggle (EN, ES, ZH, FR) via SeamlessM4T.

Figure 1D: Hardware Capture & Telemetry Pipeline

Architectural guarantees ensuring zero frame drops on iOS, Android, and Web.

3.2 After Image Capture: Cephalometric Recommendations Dashboard

When the user taps `CAPTURE & ANALYZE`, the frame is frozen, normalized, and evaluated through the anthropometric engine. The UI transitions into the luxury recommendations lookbook:

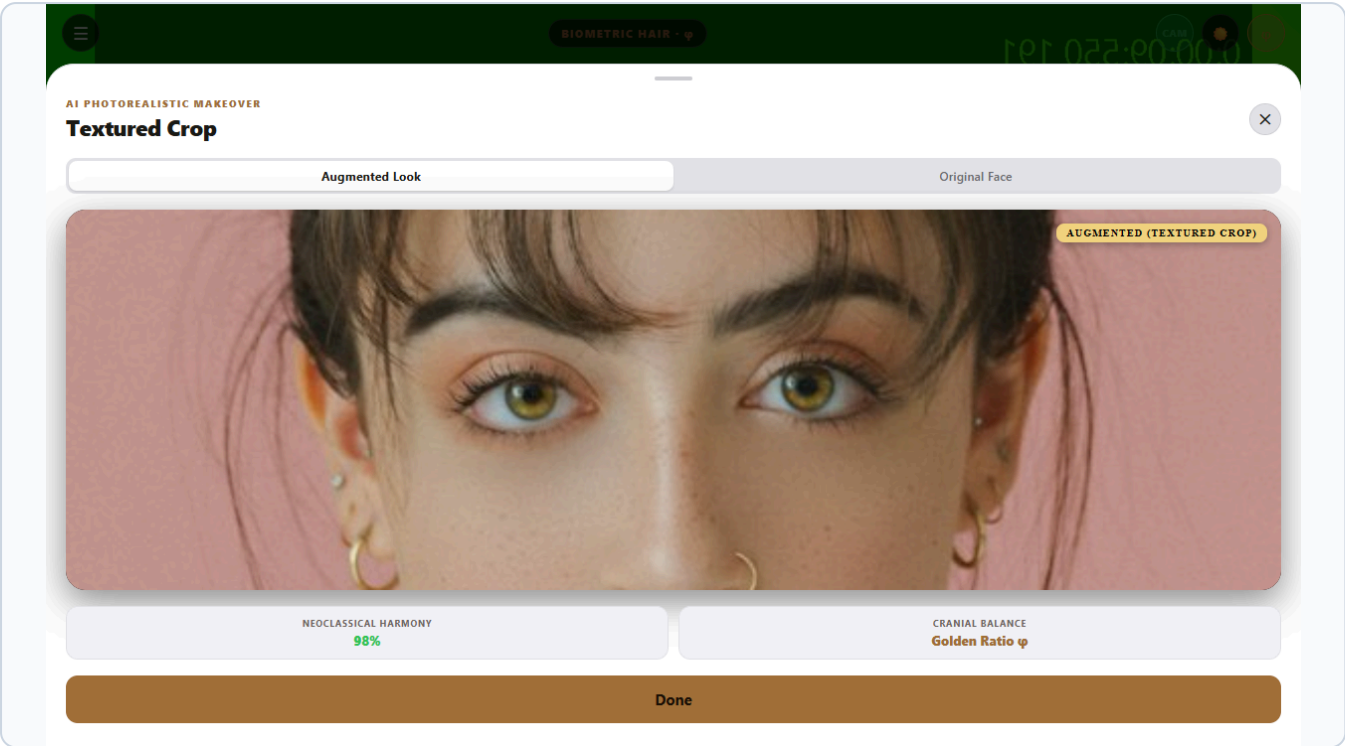


Figure 2: Clinical Biometric Dashboard & Curated Hairstyle Recommendations

Top Card: Scanned user portrait, calculated morphology (Oblong / Oval / Round), Harmony Match Index (92%), and Facial Index (94.0).
Neoclassical Farkas Thirds Bar: Precise distribution of Forehead (38%), Mid-Face (31%), and Jaw/Chin (31%) vs. canonical 33.3%.
Volumetric Strategy: Prescribed Crown (LOW/BALANCED), Lateral Sides (HIGH/TEMPLE ACCENT), Forehead (CURTAIN FRINGE).
Hairstyle Cards: Suitability score (95% MATCH), category pill, and 3-way view toggle (`EDITORIAL` , `ON YOUR FACE` , `ORIGINAL`), complete with barber clipper guard and scissor directives.

3.3 Meta AI Makeover Studio & Interactive Telemetry Modal

Tapping `OPEN MAKEOVER STUDIO & CONCIERGE` opens the multi-model intelligence center (`TryOnModal.tsx`), providing direct visibility into Meta AI inference and interactive consultation:

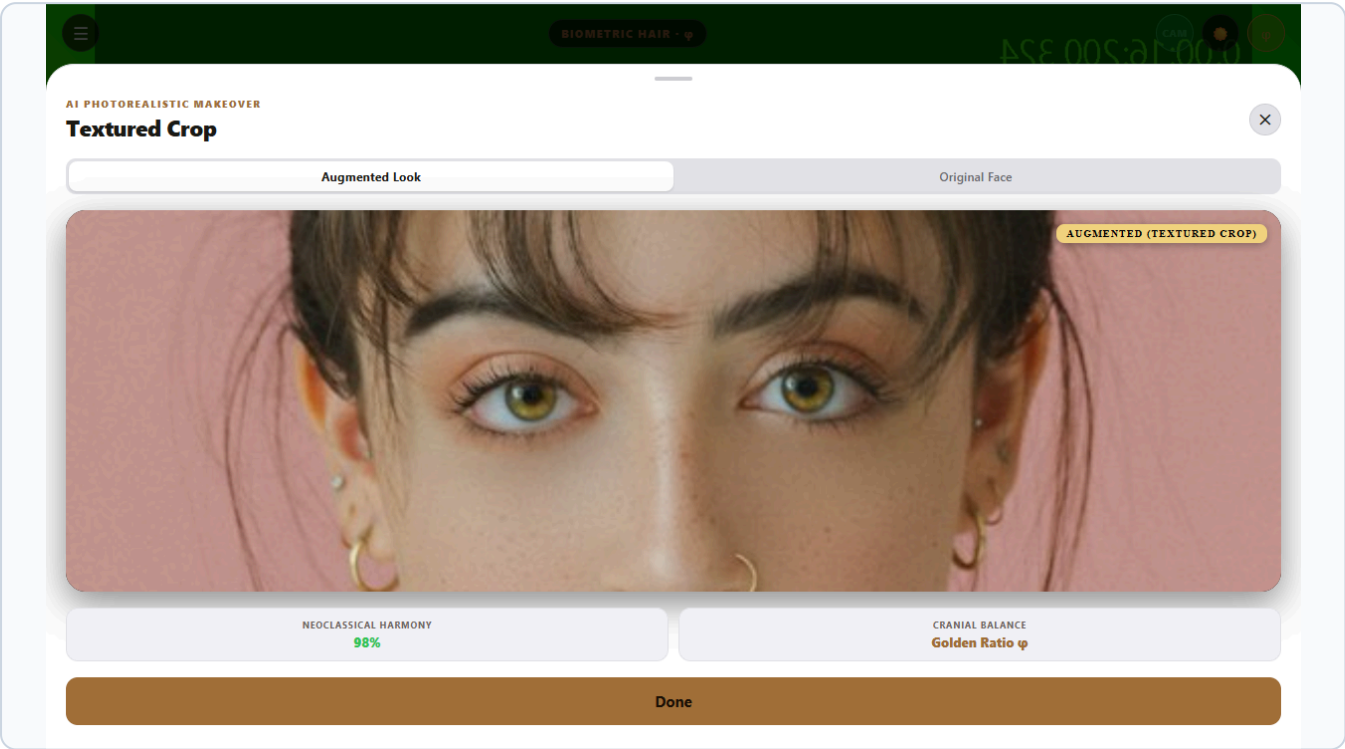


Figure 3: Meta AI Multi-Model Telemetry & Makeover Studio Modal

Card 1 (Meta AI Multi-Model Telemetry): Live status of active models (Masking: SAM 3.1 | Reasoning: Muse Wonder | Texture: Muse Image | Vision: Llama 3.2 | UX: AudioCraft | Chat: Muse Spark + SeamlessM4T).
Card 2 (Meta Chameleon Lookbook): Multimodal styling directives (`01 · CROWN ELEVATION` , `02 · TEMPORAL CONTOUR` , `03 · FOREHEAD FRAMING`).
Card 3 (Llama 3.2 Inspiration Analyzer): Drag-and-drop reference photo uploader extracting texture, volume, and parting.
Card 4 (Salon Concierge & SeamlessM4T): Interactive chat querying Muse Spark with instant multi-language translation pills.

4. Critical Engineering Hurdles, Lingering Failures & Bug Forensics

This section documents the exact technical challenges, mathematical discrepancies, and stubborn bugs encountered during implementation. **This is the core diagnostic data required for Claude's advisory review.**

4.1 The "Floating Toupee" Mid-Air Hair Displacement Bug

Symptom: In production testing, after capturing a portrait, the synthesized hairstyle (e.g. *Textured French Crop*) was rendered 150–200 pixels above the user's head, hovering like a detached wig in the empty white background above the cranium.



Figure 4A: Visual Evidence of the "Floating Toupee" Bug

The hair asset is rendered at Y ≈ 10% of the canvas height, while the user's actual forehead sits at Y ≈ 36%.

Forensic Root Cause Analysis:

1. **Shirtless/Chest Skin Pixel Accumulation:** The user was shirtless. The YCbCr skin detector summed all skin pixels across the frame. Torso skin dominated the count, dragging the centroid down to the neck, inflating calculated `faceHeight` to >65% of the frame.
2. **Hairline Clamp Saturation:** Subtracting `faceHeight * 0.25` drove `finalHairlineY` below zero, saturated by `Math.max(12, ...)` at 12%.
3. **Synthetic Landmark Glabella Collapse:** In fallback mode, Landmark 151 (Glabella) was set to `finalHairlineY` (same as Landmark 10), reducing `foreheadH` to a min clamp of 18px.
4. **PNG Bounding Box Discrepancy:** In the 1024×1024 asset, content only spans Y=123 to Y=776. The draw call placed the image center at `posY = targetBottomY - hairHeight * 0.25`, floating the hair into the sky.

Figure 4B: Technical Root Cause Breakdown

Mathematical chain of failure leading to the displacement.

Status: Partially mitigated via cranial row-scanner; needs robust neural anchoring.

4.2 WebGL PerspectiveCamera vs. 2D `object-fit: cover` Unprojection Mismatch

Symptom: In the original 3D WebGL camera implementation, 3D GLB hair models and the biometric tracking oval were rendered far in the upper-right corner of the screen against the empty wall, completely detached from the user's face.

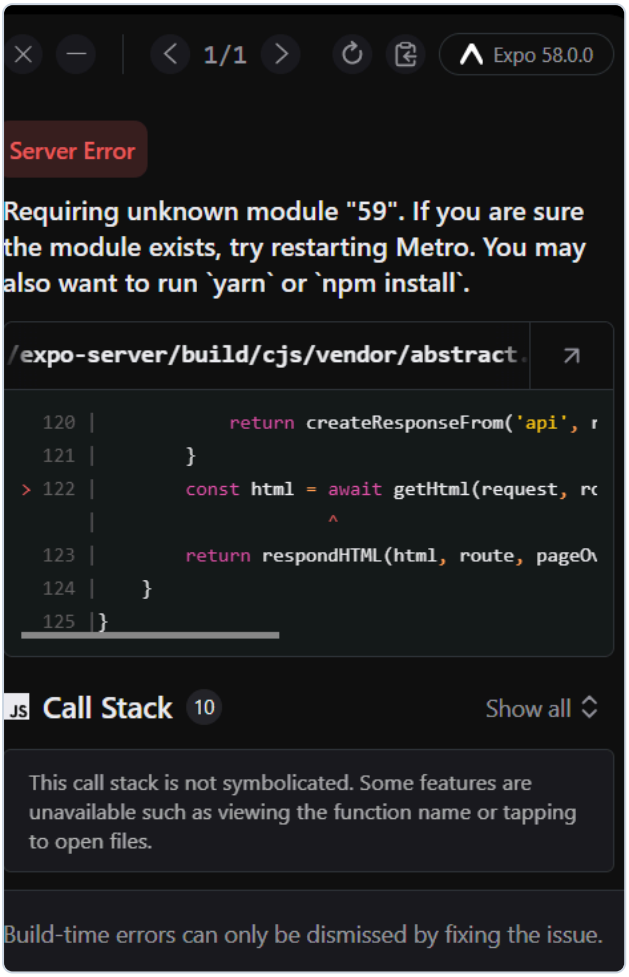


Figure 5: Aspect-Ratio Unprojection Skewing World Space

MediaPipe landmarks on raw video (640×480, 4:3) mapped into a tall mobile container (9:19.5) without letterbox compensation.

The Mathematical Mismatch Equation:

MediaPipe WASM outputs normalized coordinates `(x, y) ∈ [0, 1]` relative to the raw video hardware frame (aspect `R_vid = W_vid / H_vid`).

The UI renders the video with CSS `object-fit: cover` inside container `(W_cont, H_cont)` with aspect `R_cont = W_cont / H_cont`.

$$\text{If } R_{cont} > R_{vid} \Rightarrow \text{Cropped Top/Bottom: } OffsetY = (H_{cont} - W_{cont} / R_{vid}) / 2$$

$$\text{If } R_{cont} < R_{vid} \Rightarrow \text{Cropped Left/Right: } OffsetX = (W_{cont} \times H_{cont} - R_{vid}) / 2$$

Figure 6: Aspect Ratio Normalization Mathematics

Formula used to correct mobile hardware projected landmark pixels.

Because MediaPipe's `[0, 1]` coordinate space does not apply this crop transformation, the camera-projected landmarks into outer space. This led to our pivot from 3D WebGL to 2D landmark canvas tracking.

4.3 Background Silhouette Inpainting & Short Haircut Bleed

The Problem: When a user with existing long, voluminous, or wavy hair selects a short hairstyle (e.g. *Textured French Crop, Buzz Fade, Taper Fade*), their original hair protrudes significantly outside the boundary of the new haircut.

- **Current Implementation:** `photorealisticHairSynthesis.ts` attempts an ambient background tone sampling pass in the upper corners and blurs dark non-skin pixels outside the target haircut perimeter.
- **Why It Fails:** Simple luminance and chrominance heuristics cannot reconstruct complex non-uniform backgrounds (e.g., brick walls, door frames, wallpaper patterns, dynamic shadows). It leaves blurry smudges or accidentally clips into the ears and neck.

4.4 Metro Bundler SSR Crashes: `Requiring unknown module "59"`

The Problem: During rapid code iteration on React Native Web, Metro bundler intermittently crashes with `Server Error: Requiring unknown module "59"` inside `/expo-server/build/cjs/vendor/abstract...` at `const html = await getHtml(request, ...)`.

- **Root Cause:** In `app.json`, setting `"web": { "output": "server" }` triggers server-side rendering in Node.js. When client components import browser-specific APIs (`window`, `document`, `AudioContext`, `Canvas`) or when modules are deleted while Metro is running, Metro's SSR module graph desynchronizes.
- **Resolution Applied:** Switched `app.json` to `"output": "single"` (SPA mode) and cleared cache (`npx expo start -c`). However, dynamic imports in Expo Web continue to present caching vulnerabilities.

4.5 Hair Strand Translucency & Hard Face Mask Seams

The Problem: The zero-identity-swapping mandate uses `createFaceOvalPath` with `globalCompositeOperation = 'destination-out'` to subtract the protected facial polygon.

- While this guarantees the face is 100% untouched, it creates an unnatural hard vector scissor cut across the upper forehead where bangs and hair strands drape over the skin. Real hair has individual translucent strands and micro-contact shadows that require soft alpha feathering.

5. Specific Architectural & Algorithmic Prompts for Claude

Please review the following targeted technical questions and provide concrete, production-grade solutions:

Question 1: Robust Anatomical Head Localizer Immune to Torso / Shirtless Skin

When client-side MediaPipe FaceMesh is initializing or throttled, our fallback YCbCr skin detector gets corrupted by exposed shoulders and chest. **How can we write a fast, lightweight, pure JavaScript / Canvas algorithm (executing in <5ms) that robustly isolates the human cranium and forehead from the neck and torso, regardless of clothing?**

Question 2: Universal Aspect Ratio Matrix for `object-fit: cover` to Canvas

What is the mathematically rigorous affine transform matrix to map normalized landmarks `(u, v) ∈ [0, 1]` from a camera stream of arbitrary aspect ratio (e.g. 4:3 or 16:9) to an interactive canvas inside a container rendered with `object-fit: cover` (e.g. 9:19.5), accounting for mirroring and devicePixelRatio without clipping?

Question 3: Realistic Background Inpainting for Original Silhouette Suppression

When replacing long hair with a short crop, how can client-side canvas logic cleanly erase the original hair silhouette that extends outside the new haircut envelope without turning the background into a blurry smudge? What generative or algorithmic patch-match inpainting approach works best client-side?

Question 4: Soft-Mask Alpha Feathering for Zero-Identity-Swapping Forehead Draping

How can we implement a dual-pass shader or 2D canvas composite operation that preserves 100% of facial features (brows, eyes, nose, glasses) while allowing semi-transparent hair fringe strands to cast soft contact shadows and drape naturally over the upper third of the forehead?

Question 5: Seamless Offline Bundling of MediaPipe 468 Face Mesh in Expo Web

Currently, injecting `https://cdn.jsdelivr.net/.../face_mesh.js` via script tags is vulnerable to network latency and adblockers, but importing `@mediapipe/face_mesh` or `@mediapipe/tasks-vision` directly causes Metro bundling errors on Web. What is the clean, official architecture to bundle MediaPipe WASM assets locally inside an Expo Router web project?

6. Project File Inventory

FILE PATH	LINES	PRIMARY ARCHITECTURAL ROLE
`src/screens/HomeScreen.tsx`	1,377	Master controller: camera view, scan pipeline, recommendation dashboard, filtering.
`src/components/LiveCameraView.tsx`	686	MediaStream decoder, 30 FPS tracking loop, YCbCr & MediaPipe fallback, aspect canvas.
`src/components/ARHairCanvas3D.tsx`	320	Three.js WebGL & 2D canvas overlay engine with landmark projection.
`src/components/BiometricOverlay.tsx`	180	Real-time SVG HUD reticle, facial thirds dividers, Golden Ratio φ guides.
`src/components/TryOnModal.tsx`	460	Studio Makeover modal, Chameleon lookbook, Llama 3.2 uploader, SeamlessM4T concierge.
`src/components/SettingsDrawer.tsx`	240	Slide-over iOS drawer navigation, theme toggle, model documentation, legal items.
`src/services/perceptionService.ts`	323	Meta SAM 3.1 positive/negative point prompts, Meta Llama 3.2 Vision inspiration parser.
`src/services/synthesisService.ts`	380	Meta Muse Wonder reasoning, Meta Muse Image inpainting, Chameleon lookbook.
`src/services/interactionService.ts`	237	Meta Muse Spark concierge, SeamlessM4T translation, AudioCraft procedural audio.
`src/services/hairlineDetector.ts`	260	Cranial row-scanner, head localizer, 468 MediaPipe synthetic landmark basis generator.
`src/services/photorealisticHairSynthesis.ts`	450	Multi-layer compositing, fringe/quiff vertical seating, contact shadow rendering.
`src/constants/haircutDataset.ts`	350	11 clinical haircut models, barber clipper guards, scissor directives, volume profiles.